

Short-answer key

1. **Why subtract the max before softmax?**

To prevent overflow in `exp(logits)`. Softmax is unchanged by adding or subtracting the same constant from every class logit in a row.

2. **Why divide by batch size?**

Because the loss is the mean over examples. The gradient of a mean should also be scaled by the number of examples in the batch.

3. **Why use the ReLU mask?**

ReLU has derivative 1 where the pre-activation is positive and 0 where it is negative, so gradients only flow through active units.

4. **Why compare with autograd?**

It isolates whether the bug is in your math or elsewhere. If manual and autograd gradients match, your backward pass is almost certainly correct.

5. **Why inspect highest-loss examples?**

They are often mislabeled, outliers, or samples where the model is very confidently wrong. Those are strong candidates for data-quality review.